

# State of Developer Ecosystem 2024

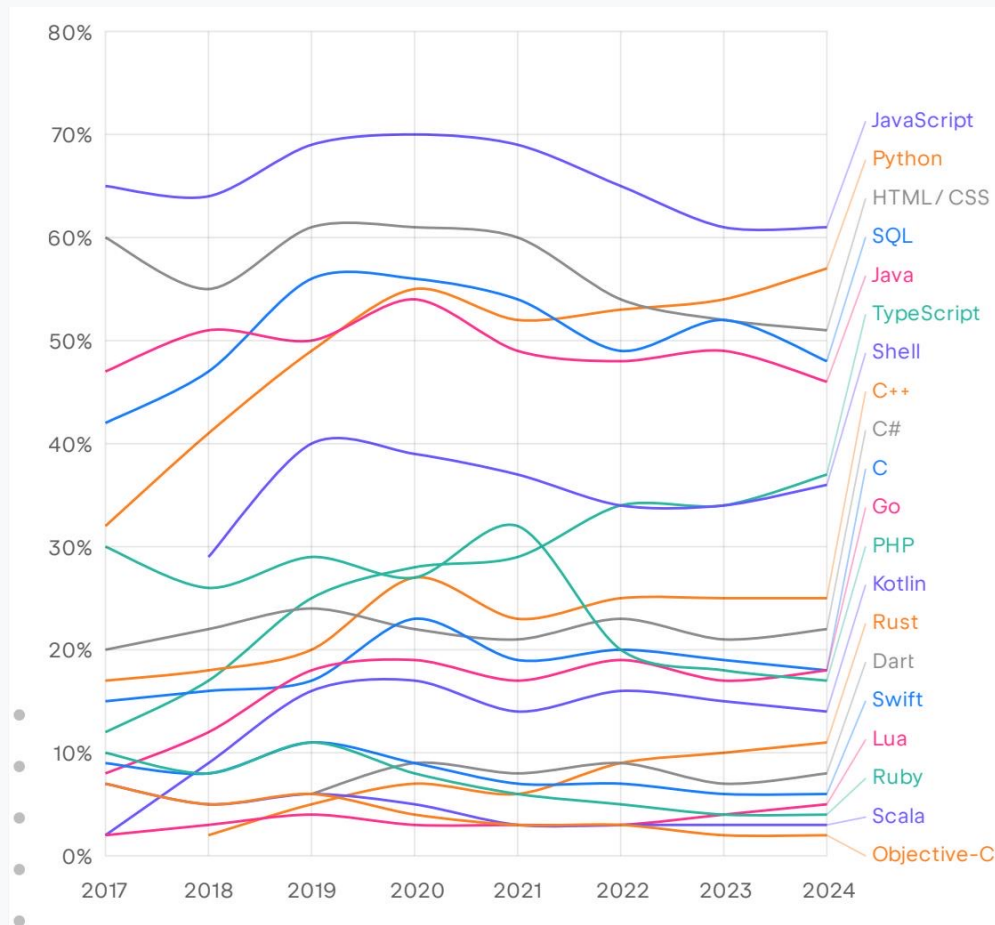
## Language Trends, Satisfaction, and Strategic Shifts

Key insights from JetBrains' survey of 23,262 developers worldwide (2017–2024)

### This deck answers

- How usage is changing
- Domain fit by language
- What developers prefer
- What to do in 2025

# Python Closes the Gap: 8 Years of Language Usage Shifts



Usage share across languages (2017–2024)

## Headline shifts

- JavaScript: 65% (2017) → 61% (2024)  
Peak 70% in 2020
- Python: 32% (2017) → 57% (2024)  
Steady growth
- TypeScript: 12% (2017) → 37% (2024)  
Tripled adoption
- PHP: 30% (2017) → 17% (2024)  
Long-term decline

## Implication

The “top-5” is reshuffling.  
Usage momentum follows domains  
(AI/data + modern tooling).

# Python Is Now the #1 Primary Language at 35%

## Primary language share

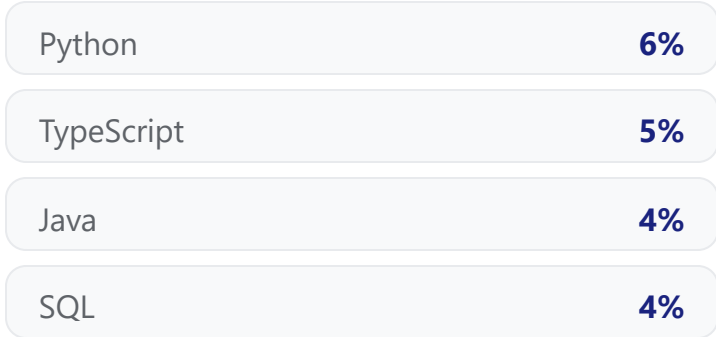


Primary language = the main language used most often

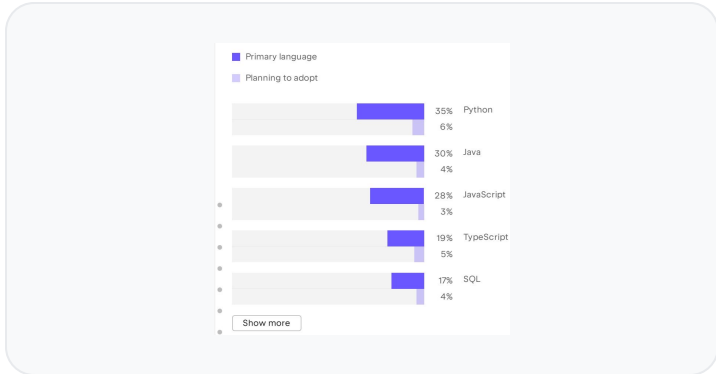
### Comfort zone effect

Planning to adopt a new primary language stays below 6%.  
Stacks are sticky once teams invest in tooling and expertise.

## Planning to adopt (primary)



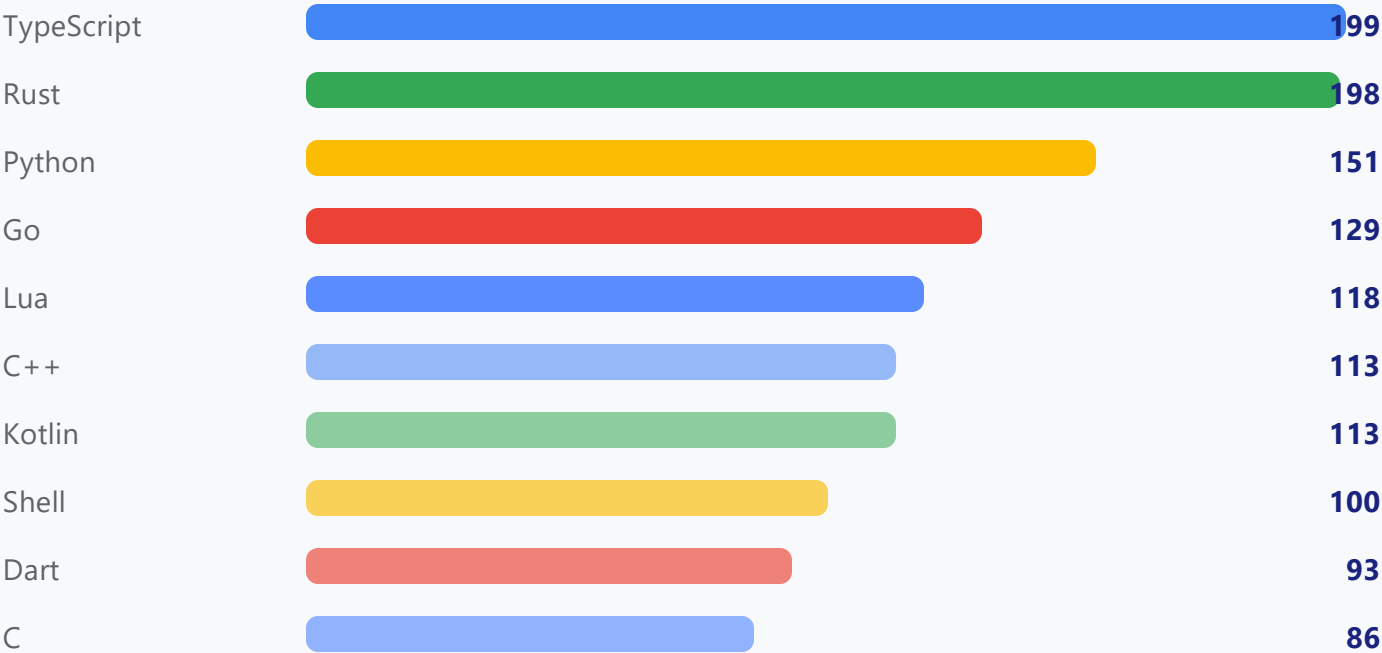
Adoption intent remains modest across languages.



Reference: Primary vs planning to adopt

# TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198

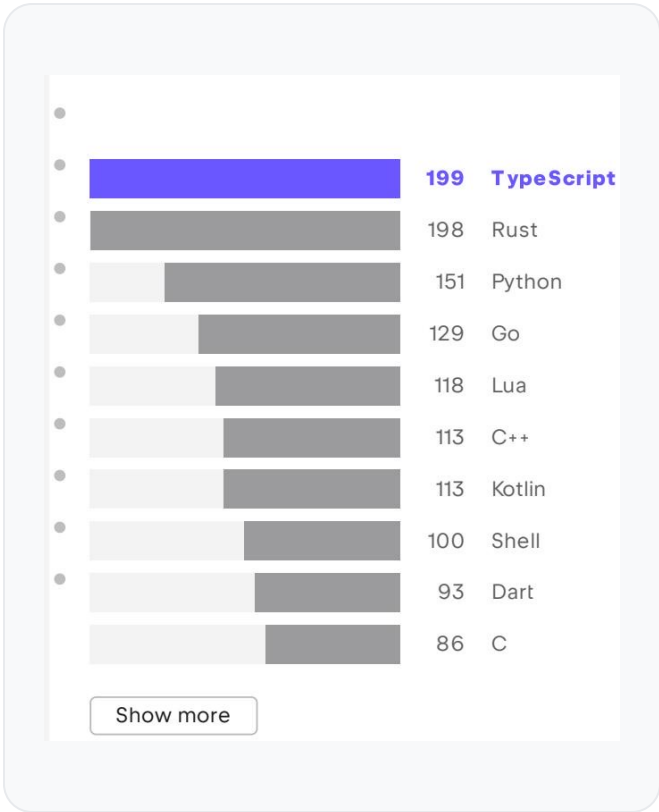
## Satisfaction score (top 10)



Higher = more satisfaction (JetBrains score)

Insight: type-safe, modern-tooling languages dominate the top.

## Reference



Source chart from the brief

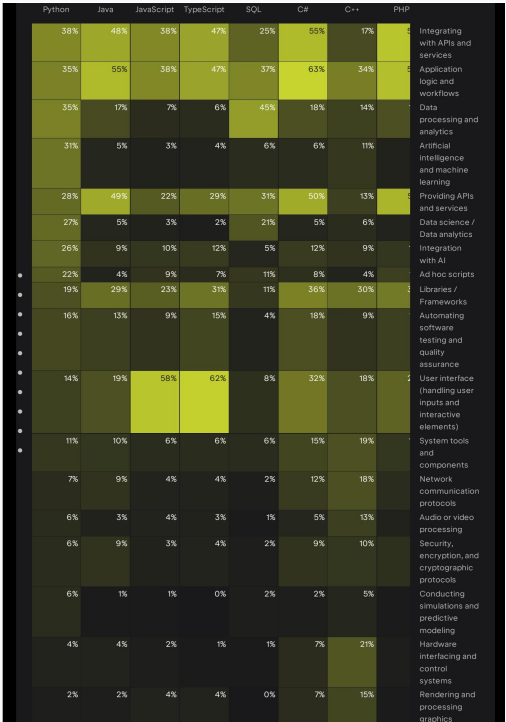
# Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

## Domain leaders (share selecting top language)

| Domain                           | Top language | Share |
|----------------------------------|--------------|-------|
| AI / Machine Learning            | Python       | 31%   |
| Data science / data analytics    | Python       | 28%   |
| Data processing & analytics      | Python       | 17%   |
| Integrating with APIs & services | Python       | 38%   |
| User Interface                   | TypeScript   | 62%   |
| APIs & Services                  | Java         | 49%   |
| Application logic & workflows    | Java         | 55%   |
| System tools & components        | C++          | 27%   |

Note: UI also shows JavaScript at 58% in the source brief.

## Heatmap (reference)



Source heatmap from the brief

# What This Means for Your Technology Strategy in 2025

## Actions for engineering leaders

### 1) Invest in Python for AI/ML

Python leads AI/ML (31%) and data science (28%).  
Make it a first-class hiring + tooling track.

### 2) Standardize TypeScript for UI

TypeScript dominates UI (62%) and tops satisfaction (199).  
Treat it as the default for frontend/full-stack.

### 3) Rust + Go for infrastructure

Rust (198) and Go (129) combine performance and safety.  
Go is widely used (~18%); Rust is climbing (11%).

### 4) Plan for shrinking pools

PHP declines from 30% (2017) to 17% (2024).  
Ruby declines from 10% (2017) to 4% (2024).